



S. J. P. N. TRUST'S
HIRASUGAR INSTITUTE OF TECHNOLOGY, NIDASOSHI

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Department of Computer Science & Engineering

MACHINE LEARNING

Module 3:

Chapter 5: Regression Analysis

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Introduction to Regression

- Regression analysis is the premier method of supervised learning.
- This is one of the most popular and oldest supervised learning technique.
- Given a training dataset D containing N training points (x_i, y_i) , where $i=1\dots N$, regression analysis is used to model the relationship between one or more independent variables x_i ; and a dependent variable y_i .
- The relationship between the dependent and independent variables can be represented as a function as follows: **$y=f(x)$** .
- Here, the feature variable x is also known as an explanatory variable, exploratory variable, a predictor variable, an independent variable, a covariate, or a domain point.
- y is a dependent variable.
- Dependent variables are also called as labels, target variables, or response variables.

Introduction to Regression

- **Regression analysis determines the change in response variables when one exploration variable is varied while keeping all other parameters constant.**
- This is used to determine the relationship each of the exploratory variables exhibits. Thus, **regression analysis is used for prediction and forecasting.**
- Regression is used to predict continuous variables or quantitative variables such as price and revenue.

Introduction to Regression

- Thus, the primary concern of regression analysis is to find answer to questions such as:
 1. What is the relationship between the variables?
 2. What is the strength of the relationships?
 3. What is the nature of the relationship such as linear or non-linear?
 4. What is the relevance of the attributes?
 5. What is the contribution of each attribute?
- There are many applications of regression analysis. Some of the applications of regressions include predicting:
 1. Sales of a goods or services
 2. Value of bonds in portfolio management
 3. Premium on insurance companies
 4. Yield of crops in agriculture
 5. Prices of real estate

- The quality of the regression analysis is determined by the factors such as **correlation and causation(Interconnection/relationship).**

Regression and Correlation

- Correlation among two variables can be done effectively using a **Scatter plot**, which is a plot between explanatory variables and response variables.
- It is a 2D graph showing the relationship between two variables. The x-axis of the scatter plot is independent, or input or predictor variables and y-axis of the scatter plot is output or dependent or predicted variables.
- The scatter plot is useful in exploring data.
- The **Pearson correlation coefficient is the most common test for determining correlation** if there is an association between two variables. The correlation coefficient is denoted by r .
 - **In positive correlation**, one variable change is associated with the change in another variable.
 - **In negative correlation**, the relationship between the variables is reciprocal.
 - **In random correlation**, no relationship exists between variables.

Regression and Correlation

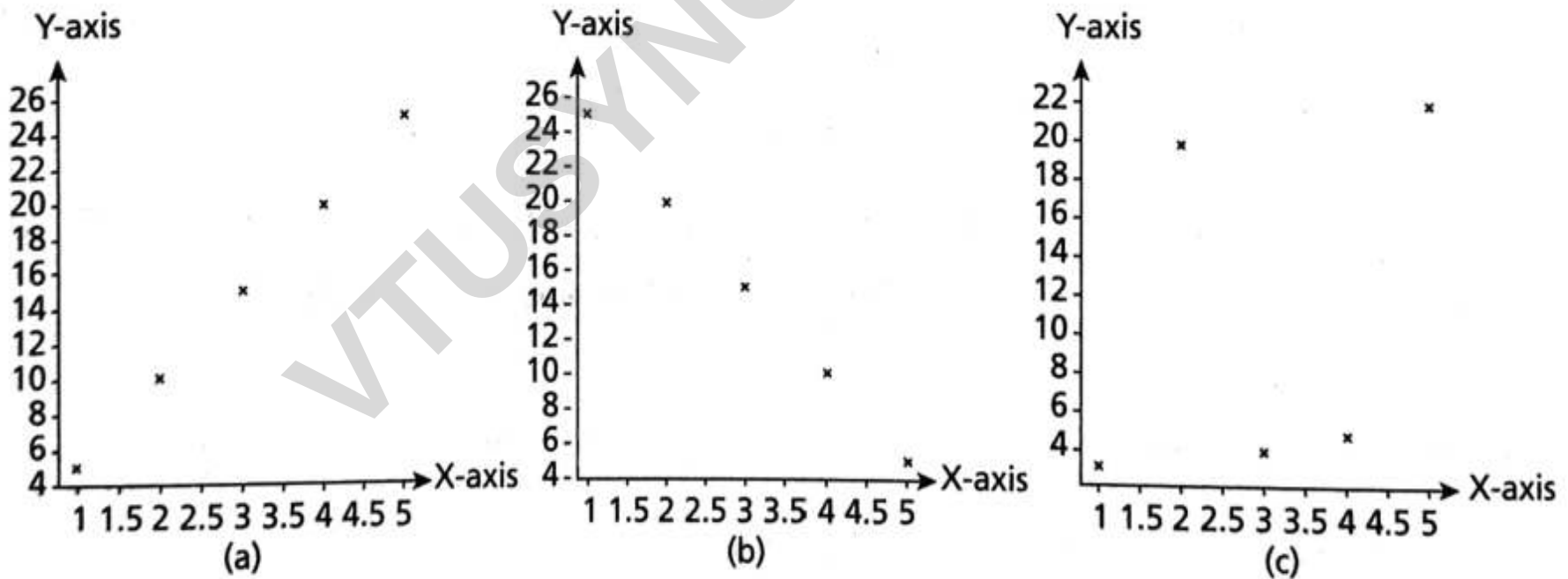


Figure 5.1: Examples of (a) Positive Correlation (b) Negative Correlation (c) Random Points with No Correlation

Regression and Causation

- Causation is about **causal relationship among variables, say x and y.**
- Causation means knowing **whether x causes y to happen or vice versa. x causes y is often denoted as x implies y. Correlation and Regression relationships are not same as causation relationship.**
- For example, the correlation between economical background and marks scored does not imply that economic background causes high marks.
- Similarly, the relationship between higher sales of cool drinks due to a rise in temperature is not a causal relation. Even though high temperature is the cause of cool drinks sales, it depends on other factors too.

Linearity and Non-linearity Relationships

- The linearity relationship between the variables means the **relationship between the dependent and independent variables can be visualized as a straight line.**
- The line of the form, **$y = ax + b$**
- can be fitted to the data points that indicate the relationship between x and y.
- By linearity, it is meant that as one variable increases, the corresponding variable also increases in a linear manner.

Linearity and Non-linearity Relationships

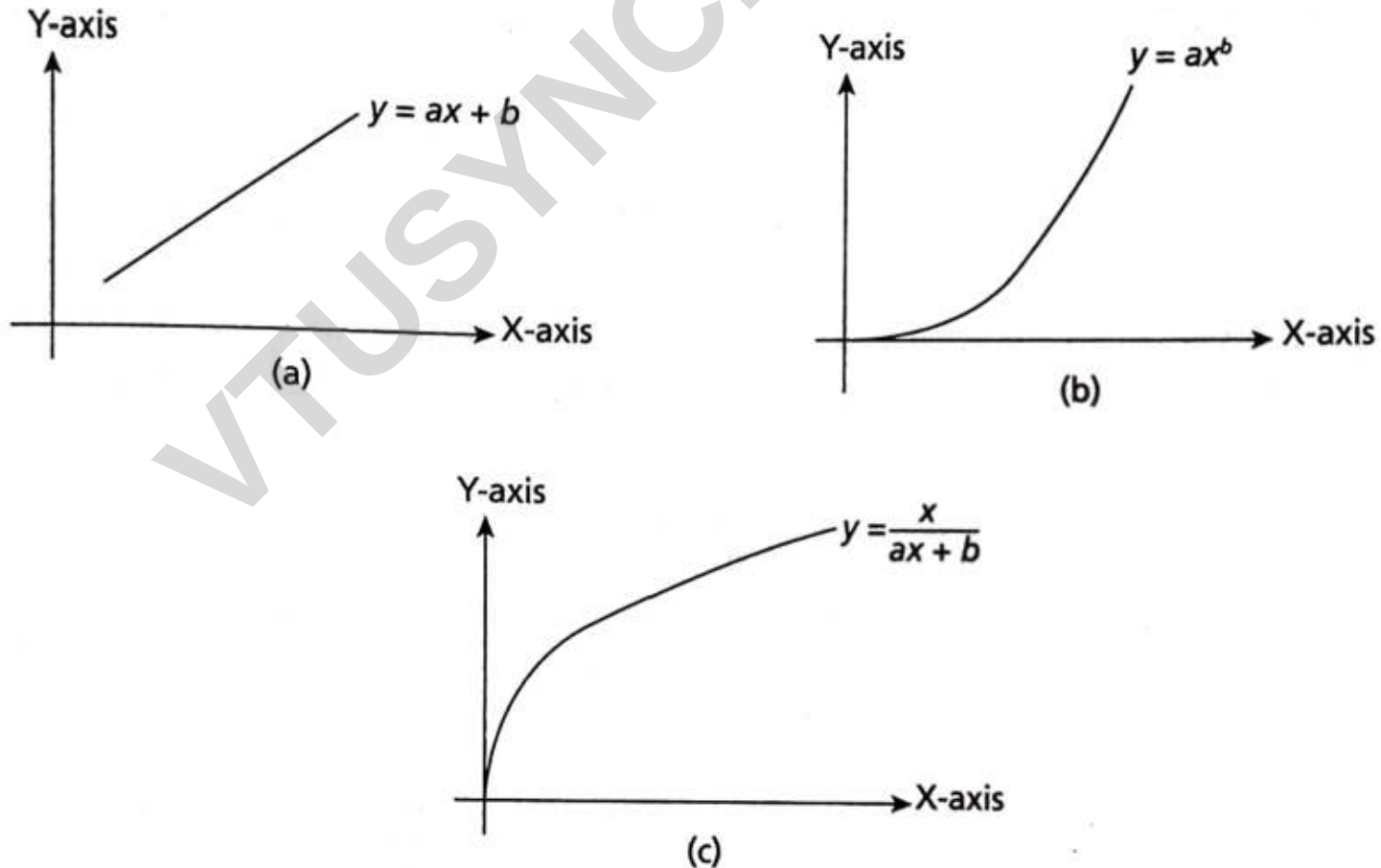
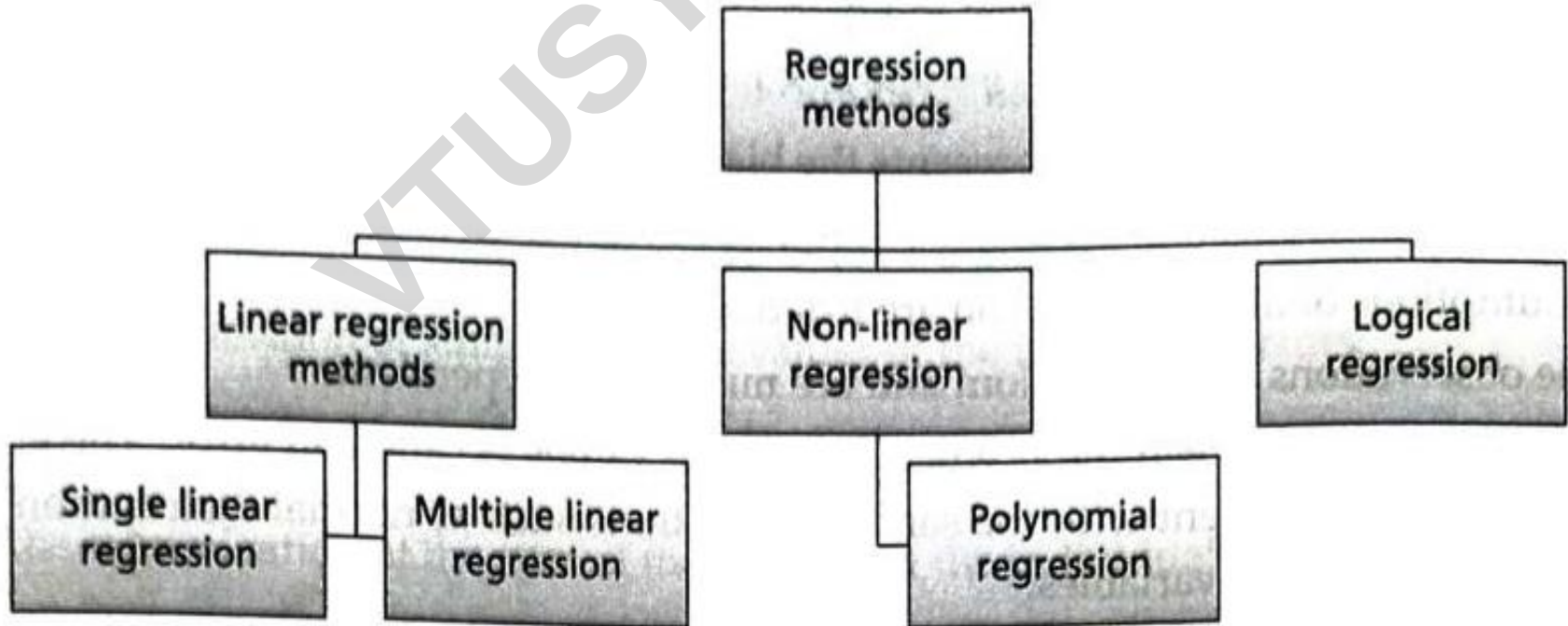


Figure 5.2: (a) Example of Linear Relationship of the Form $y = ax + b$ (b) Example of a Non-linear Relationship of the Form $y = ax^b$ (c) Examples of a Non-linear Relationship $y = \frac{x}{ax + b}$

Types of Regression Methods



Types of Regression Methods

- **Linear Regression** It is a type of regression where a line is fitted upon given data for **finding the linear relationship between one independent variable and one dependent variable** to describe relationships.
- **Multiple Regression** It is a type of regression where a line, is fitted for **finding the linear relationship between two or more independent variables and one dependent variable** to describe relationships among variables.
- **Polynomial Regression** It is a type of non-linear regression method of describing **relation ships among variables where N th degree polynomial is used to model the relationship between one independent variable and one dependent variable**. Polynomial multiple regression is used to model two or more independent variables and one dependant variable.
- **Logistic Regression** It is used for **predicting categorical variables that involve one or more independent variables and one dependent variable**. This is also known as a binary classifier.
- **Lasso and Ridge Regression Methods** These are special variants of regression method where **regularization methods are used to limit the number and size of coefficients of the independent variables**.

Limitations of Regression Method

- 1. Outliers** — Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
- 2. Number of cases** — The ratio of independent and dependent variables should be at least 20 : 1.
- 3.** For every explanatory variable, there should be at least 20 samples. At least five samples are required in extreme cases.
- 4. Missing data** — Missing data in training data can make the model unfit for the sampled data.
- 5. Multicollinearity** — If exploratory variables are highly correlated (0.9 and above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1.

LINEAR REGRESSION

- the linear regression model can be **created by fitting a line among the scattered data points**. The line is of the form given in Eq. (5.2). $y = a_0 + a_1 \times x + e$
- Here, a_0 is the intercept which represents the bias and a_1 represents the slope of the line. These are called regression coefficients. e is the error in prediction.
- **The assumptions of linear regression are listed as follows:**
 - The observations (y) are random and are mutually independent.
 - The difference between the predicted and true values is called an error.
 - The distribution of the error term is independent of the joint distribution of explanatory variables.
 - The unknown parameters of the regression models are constants.

LINEAR REGRESSION

- The idea of linear regression is based on **Ordinary Least Square (OLS) approach**.
- In this method, the **data points are modelled using a straight line**.

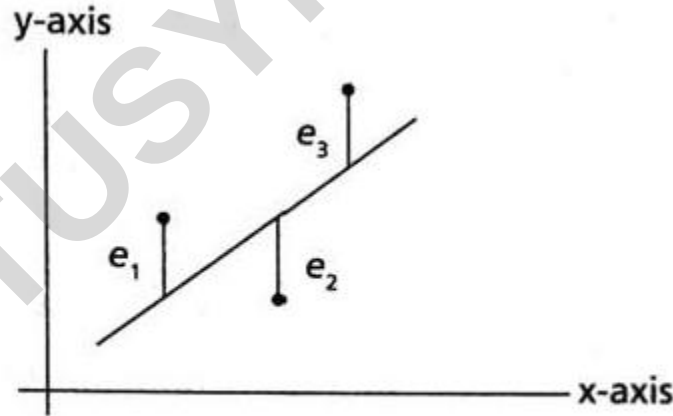


Figure 5.4: Data Points and their Errors

- In Figure 5.4, three data points and their errors (e_1, e_2, e_3) are shown.
- The vertical distance between each point and the line predicted by the approximate line equation $y = a_0 + a_1x$ is called an error.
- These individual errors are added to compute the total error of the predicted line. This is called sum of residuals.
- The squares of the individual errors can also be computed and added to give a sum of squared error.
- The line with the lowest sum of squared error is called line of best fit.

LINEAR REGRESSION

- Mathematically, based on Eq. (5.2), the line equations for points $(x_1, x_2, \dots, x_n)$ are:

$$y_1 = (a_0 + a_1x_1) + e_1$$

$$y_2 = (a_0 + a_1x_2) + e_2$$

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$$y_n = (a_0 + a_1x_n) + e_n$$

- In general, the error is given as $e_i = y_i - (a_0 + a_1x_i)$
- Here, the terms $(e_1, e_2, \dots, e_n)$ are error associated with the data points and denote the difference between the true value of the observation and the point on the line.
- This is also called as residuals.
- The residuals can be positive, negative or zero.

LINEAR REGRESSION

- A regression line is the line of best fit for which the sum of the squares of residuals is minimum.
- The minimization can be done as minimization of individual errors by finding the parameters a_0 and a_1 , such that:

$$E = \sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - (a_0 + a_1 x_i)) \quad (5.5)$$

Or as the minimization of sum of absolute values of the individual errors:

$$E = \sum_{i=1}^n |e_i| = \sum_{i=1}^n |(y_i - (a_0 + a_1 x_i))| \quad (5.6)$$

Or as the minimization of the sum of the squares of the individual errors:

$$E = \sum_{i=1}^n (e_i)^2 = \sum_{i=1}^n (y_i - (a_0 + a_1 x_i))^2 \quad (5.7)$$

- Therefore, linear regression is modelled as a minimization function as follows:

LINEAR REGRESSION

- Therefore, linear regression is modelled as a minimization function as follows:

$$\begin{aligned} J(a_1, a_0) &= \sum_{i=1}^n [y_i - f(x_i)]^2 \\ &= \sum_{i=1}^n [y_i - (a_0 + a_1 x_i)]^2 \end{aligned} \quad (5.8)$$

Here, $J(a_0, a_1)$ is the criterion function of parameters a_0 and a_1 . This needs to be minimized. This is done by differentiating and substituting to zero. This yields the coefficient values of a_0 and a_1 . The values of estimates of a_0 and a_1 are given as follows:

$$a_1 = \frac{(\overline{xy}) - (\bar{x})(\bar{y})}{(\overline{x^2}) - (\bar{x})^2} \quad (5.9)$$

And the value of a_0 is given as follows:

$$a_0 = (\bar{y}) - a_1 \times \bar{x} \quad (5.10)$$